

States of Matter

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Solids have fixed shape and volume, liquids have fixed volume but take the container shape, and gases have neither fixed shape nor fixed volume because particles are farther apart and move more freely.
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Round 1 - Retrieve It

1. Which state has a fixed shape and fixed volume?

2. Which state has a fixed volume but takes the shape of its container?

3. Which state has neither fixed shape nor fixed volume?

4. Which state usually has the greatest particle spacing?

Round 2 - Sketch It

Draw particle models for a solid, liquid, and gas in equal-size boxes. Show relative spacing and arrangement.

States of Matter - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. Why can air in a syringe be compressed more easily than water?

2. Name the change from solid to liquid and from liquid to gas.

3. Name the change from gas to liquid and from liquid to solid.

Round 4 - Reason from Evidence

Explain what happens to particle motion and spacing as matter is heated from solid to liquid to gas.

States of Matter - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain States of Matter in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect state of matter to particle motion. Why does a liquid flow while a solid keeps its shape?

E**Extend**

Predict what happens to the particles of water vapor as it cools enough to condense on a cold surface.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - States of Matter

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Solid.
2. Liquid.
3. Gas.
4. Gas.

Sketch It - look for

Solid particles close and ordered; liquid particles close but disordered/mobile; gas particles much farther apart.

Apply It

1. Gas particles have much more empty space between them.
2. Melting; vaporization/evaporation.
3. Condensation; freezing.

Reason from Evidence - look for

Particles gain kinetic energy, move faster, and become less constrained; spacing generally increases, especially in the gas state.

ACE - Extend look-for

Particles lose energy, slow down, move closer together, and form liquid water.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.